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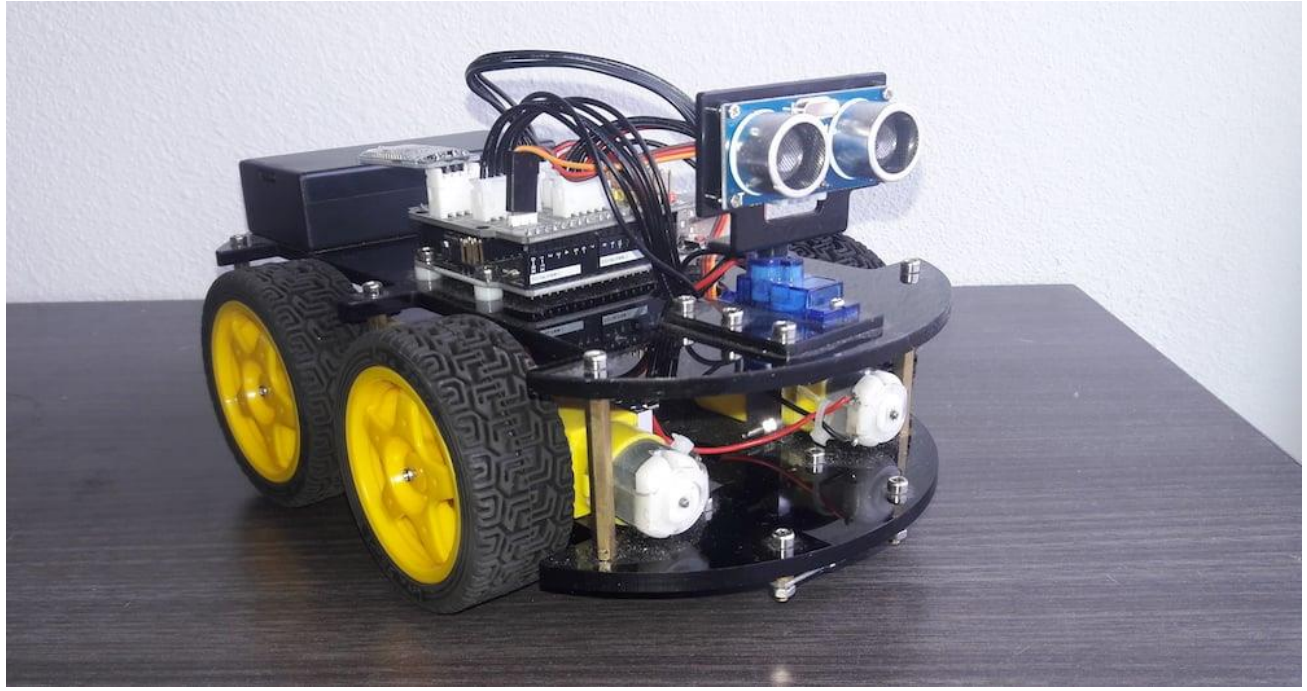
DIV:- B

PRESENTATION ON LINE FOLLOWER ROBOT



INTRODUCTION FOR LINE FOLLOWER ROBOT

- A LINE FOLLOWER ROBOT IS AN AUTONOMOUS ROBOT WHICH IS ABLE TO FOLLOW EITHER A BLACK LINE THAT IS DRAWN ON THE SURFACE CONSISTING OF A CONTRASTING COLOR.
- THE ROBOT USES INFRARED SENSORS TO IDENTIFY THE LINE, THUS ASSISTING THE ROBOT TO STAY ON THE TRACK.
- THE ROBOT IS DRIVEN BY DC GEAR MOTORS TO CONTROL THE MOVEMENT OF THE WHEELS. THE DC GEAR MOTOR IS DRIVEN BY THE MOTOR DRIVEN CIRCUIT.

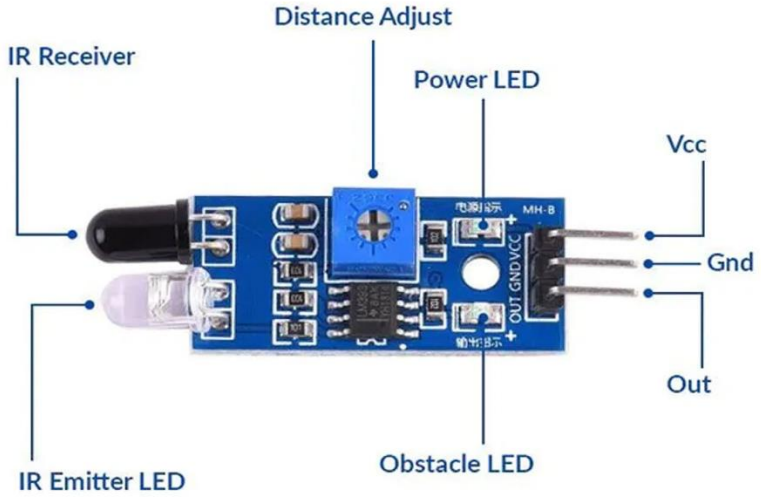
TECHNOLOGIES OF LINE FOLLOWER ROBOT

- LINE FOLLOWER ROBOTS ARE AUTONOMOUS ROBOTS DESIGNED TO FOLLOW A PATH, USUALLY A LINE DRAWN ON THE GROUND. THEY ARE COMMONLY USED IN EDUCATIONAL ROBOTICS TO TEACH PROGRAMMING, ELECTRONICS, AND ROBOTICS CONCEPTS. THE TECHNOLOGIES AND COMPONENTS INVOLVED IN BUILDING A LINE FOLLOWER ROBOT

EQUIPMENT FOR
LINE FOLLOWER ROBOT

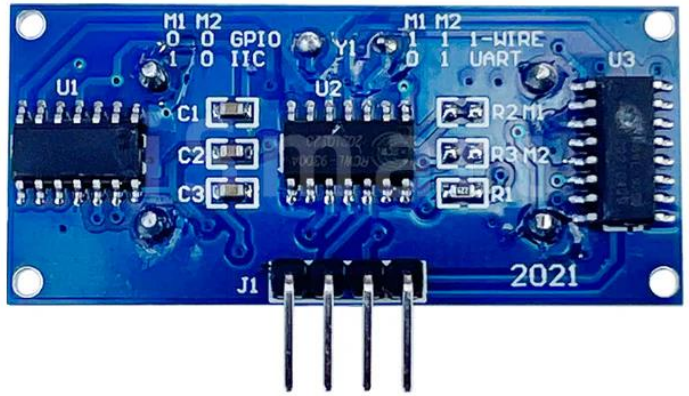
- An infrared sensor is a radiation-sensitive optoelectronic component with a spectral sensitivity in the infrared wavelength range 780n.

- IR sensor are now widely used in motion detectors, which are used in building services to switch on lamps or in alarm systems to detect unwelcome guests.

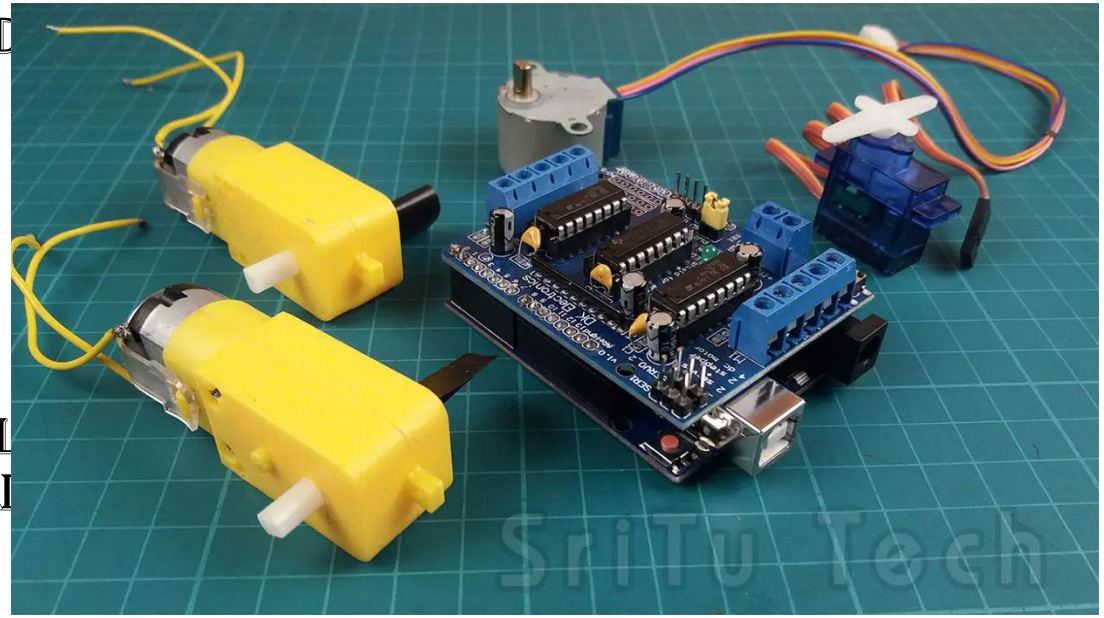


ULTRASONIC SENSOR

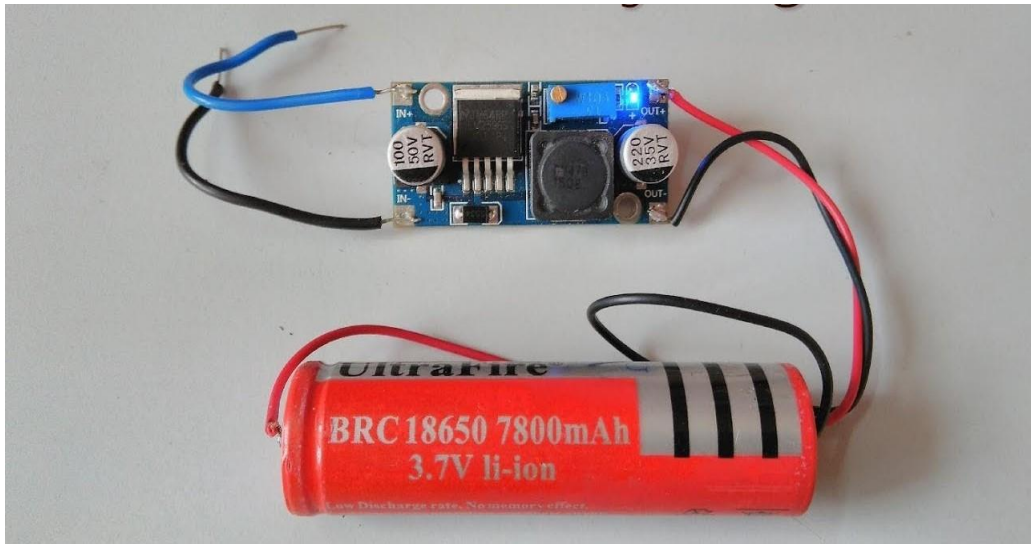
- ULTRASONIC SENSOR EMIT A CHIRP USUALLY BETWEEN 23 KHZ AND 40 KHZ MUCH HIGHER THAN THE TYPICAL AUDIBLE RANGE OF HUMAN HEARING AT 20 KHZ HENCE THE TERM ULTRASONIC
- USING THIS CHIRP THEY MEASURE THE AMOUNT OF TIME IT MAKES FOR THE SOUND TO BOUNCE OFF AN OBJECT



- THE MOTO DRIVER SHIELD MAINLY CONSISTS OF TWO MOTOR DRIVER AND ONE SHIFT REGISTER IC.
- THIS SHIELD USES THE L29D MOTOR DRIVER IC AND 74HC595 SHIFT REGIGSTR IC.
- TT GEAR MOTOR , ALSO KNOWN AS PLASTIC GEARBOX MOTORS, ARE SMALL ELECTRIC MOTORS OFTEN USED IN SMALL ROBOTICS PROJECTS AND OTHER DIY ELECTRONICS PROJECTS.



- AN 18650 BATTERY IS A RECHARGEABLE LITHIUM-ION BATTERY.
- THEY TEND TO HAVE A NOMINAL VOLTAGE OF 3.6V AND RANGE IN CAPACITY FROM 1800MAH TO 3600MAH
- A DC MOTOR IS AN ELECTRICAL MACHINE THAT CONVERT ELECTRICAL ENERGY MECHANICAL ENERGY.
- IN A DC MOTOR, THE INPUT ELECTRICAL ENERGY IS THE DIRECT CURRENT WHICH IS TRANSFORMED INTO THE MECHANICAL ROTATION



FUNCTIONALITIES FOR LINE FOLLOWER ROBOT

FUNCTIONALITY: 1

- THE PRIMARY FUNCTIONALITY OF A LINE FOLLOWER ROBOT IS TO DETECT AND FOLLOW A LINE. THIS IS USUALLY ACHIEVED USING SENSORS LIKE INFRARED (IR) OR OPTICAL SENSORS. THE SENSORS DETECT THE CONTRAST BETWEEN THE LINE (OFTEN BLACK) AND THE BACKGROUND (USUALLY WHITE OR ANOTHER CONTRASTING COLOR), ALLOWING THE ROBOT TO FOLLOW THE PATH ACCURATELY.

FUNCTIONALITY : 2

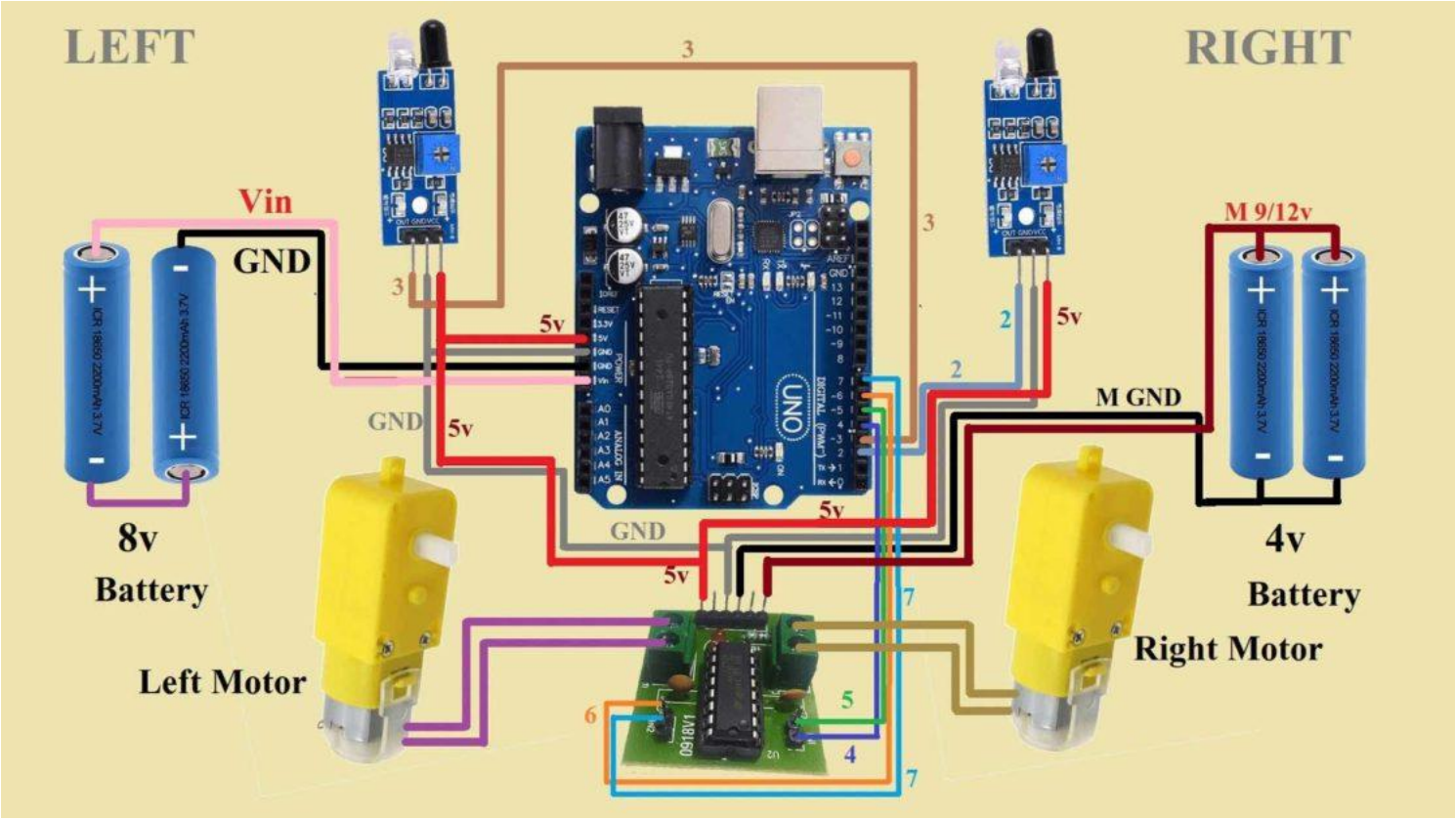
- ADVANCED LINE FOLLOWER ROBOTS MAY INCLUDE FUNCTIONALITY TO DETECT AND AVOID OBSTACLES IN THEIR PATH. THIS IS TYPICALLY DONE USING ADDITIONAL SENSORS LIKE ULTRASONIC, INFRARED, OR PROXIMITY SENSORS. WHEN AN OBSTACLE IS DETECTED, THE ROBOT CAN STOP, GO AROUND THE OBSTACLE, OR TAKE AN ALTERNATIVE PATH, DEPENDING ON ITS PROGRAMMING AND DESIGN.

FUNCTIONALITIES FOR LINE FOLLOWER ROBOT

FUNCTIONALITY : 3

- THE ROBOT NEEDS TO ADJUST ITS SPEED AND DIRECTION TO FOLLOW THE LINE ACCURATELY. THIS FUNCTIONALITY INVOLVES CONTROLLING THE MOTORS BASED ON SENSOR INPUT. FOR EXAMPLE, IF THE ROBOT DEVIATES FROM THE LINE, IT CAN ADJUST THE SPEED OF ITS WHEELS TO TURN BACK ONTO THE LINE. SPEED CONTROL CAN ALSO BE USED TO NAVIGATE CURVES OR SHARP TURNS ON THE LINE.

CIRCUIT DIAGRAM OF LINE FOLLOWER ROBOT



THANK YOU

BY.

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SARJAK PIPRIYA